

WASP-39b

R-1244

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_220331 · created 2026-07-31T11:16:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459669.9504 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.066 +/- 0.047
Transit depth (Rp/Rs)^2	0.44 +/- 0.62 [%]
Orbital Inclination (inc)	87.25 +/- 2.27
Airmass coefficient 1 (a1)	1.236 +/- 0.014
Airmass coefficient 2 (a2)	-0.1272 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	None
Optimal Aperture	4.38
Optimal Annulus	7.17
Transit Duration (day)	0.088 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.066 ± 0.047 vs 0.1457 (deviation 1.2σ)
Duration fit vs published	0.088 d vs 0.12 d (deviation 0.6σ)
Mid-time O–C	-4.6 min
Depth SNR	1.0
Residual scatter	1.15 %

Light curve

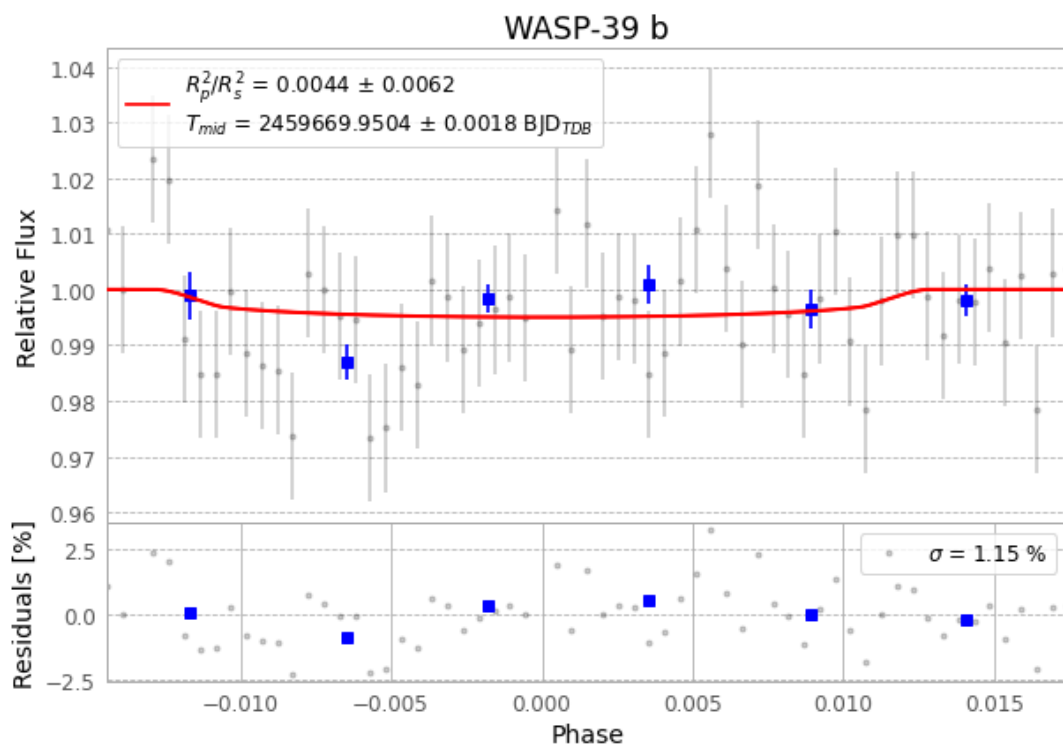


Figure 1 — FinalLightCurve_WASP-39 b_2022-03-31.png (SHA-256 f13913a0e267363f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [227, 246], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e58d8ba14accb3f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), WASP-39220331091513.FITS → WASP-39220331122113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-220331.log (b5325a422f9e...), FinalLightCurve_WASP-39 b_2022-03-31.png (f13913a0e267...), FinalLightCurve_WASP-39 b_2022-03-31.csv (dcf378618c70...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:16Z.